

Software Requirements Specification

For

Village Traveler (Smart Tourism Information System)

Version 1.0

Approved

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Revision History

Name	Date	Reason of changes	Version
M H T ALOKA	10.03.2026	Add diagrams for after get the feedback others	1.1

1. Introduction

1.1. Purpose

This document is the comprehensive Software Requirements Specification (SRS) for the "Village Tourist" Smart Tourism Information Service. It is a location-aware website service intended for tourists who wish to explore cultural, environmental and historical sites within a 25 km radius of their homestead in Sri Lanka. This document is a comprehensive description of the project scope, its features, interfaces and non-functional requirements.

1.2. Document Conventions

- **Requirement Identification:** All functional requirements shall be given unique identification using a prefix that reflects the system feature to which the requirement belongs, followed by a sequential identifier.
- **Priority:** The priority level associated with each system feature and its associated requirements shall be clearly identified as discussed in Section 4.
- **Cross-Referencing:** Requirements shall be referenced against relevant sections of the project proposal.

1.3. Intended Audience and Reading Suggestions

This document is meant for various audiences at various stages of the project. Therefore:

Project Team & Developers: This section of the audience should read this document carefully, with special attention to Section 2, Section 3, Section 4, and Section 5 of this document.

Project Supervisor & Evaluators: This section of the audience should read this document carefully, with special attention to Section 1, Section 2, and Section 4 of this document.

Testers: This section of the audience should use this document to write test cases for the project to ensure that all requirements have been met, using Section 3, Section 4, and Section 5 of this document.

This document starts with an introduction and an overall description of the project. Then it describes the external interfaces of the project and, finally, describes the functional and non-functional requirements of the project.

1.4. Product Scope

The "Village Traveler" system is a web-based application that aims to bridge the information gap for tourists visiting the Wathugedara and Balapitiya area in Sri Lanka. The main goal of the system is to enable independent, cultural, eco, and domestic tourists to visit rural attractions. This will be achieved by providing a reliable and user-friendly platform that presents information about key locations, such as photographs, descriptions, and GPS coordinates. This is done by using the user's real-time location to determine which location is closest to them. Other key features of the system are that it will be available in Sinhala and English and will integrate with Google Maps. In addition, the system will offer "One-Day Trip" plans. The end goal of the system is to promote the local economy by directing tourists away from the coastal belt and into the villages.

1.5. References

1. Aloka, M. H. T. (2026). *Smart Tourism Information System* [Unpublished student project document]. Wathugedara, Sri Lanka.
2. Amazing Lanka. (2024). *Karadeniya Shailatharama Viharaya*. [Online] Available at: [Provided in proposal]
3. Brief Garden. (2024). *Bevis Bawa's Brief*. [Online] Available at: [Provided in proposal]
4. Google. (2024). *Firebase Documentation*. [Online] Available at: [Provided in proposal]
5. Google. (2024). *Maps JavaScript API Documentation*. [Online] Available at: [Provided in proposal]
6. IEEE. (1998). *IEEE Recommended Practice for Software Requirements Specifications* (IEEE Std 830-1998).
7. Lakpura. (2024). *Ambalangoda Mask Factory and Museum*. [Online] Available at: [Provided in proposal]
8. Lanka Excursions Holidays. (2024). *Ambalangoda - Sri Lanka's village of mask manufacturers*. [Online] Available at: [Provided in proposal]
9. Lanka Excursions Holidays. (2024). *Meetiyyagoda Moonstone Mine in Sri Lanka*. [Online] Available at: [Provided in proposal]
10. Madu Ganga Foundation. (2003). *The Madu Ganga Wetland*. [Online] Available at: [Provided in proposal]
11. ResearchGate. (2021). *Android Based Application Using Google Maps API for Tourism Travel Guide*. [Online] Available at: [Provided in proposal]
12. Sri Lanka Biodiversity. (2024). *Hikkaduwa Coral Sanctuary*. [Online] Available at: [Provided in proposal]
13. Sri Lanka Tourism Development Authority. (2017). *Tourism Strategic Plan 2017 to 2020*. Colombo, Sri Lanka.

2. Overall Description

2.1. Product Perspective

The new product, "Village Traveler," is a new, self contained web product that is not a variation of an existing product but a new solution to a particular problem. The new product will function independently but will require other sources for certain functions to operate. The new product will require the GPS function of the user's device and the Google Maps service to function, and all tourism related content will be stored in a cloud database, which will be administered by the administrators of the new product.

2.2. Product Functions

The major functionalities that the "Village Traveler" system should have are:

Location Aware Attraction Discovery: Once the user accesses the system, it should identify the user's location and show a list of tourist attractions within a 25 km radius from the user's location in order from closest to farthest.

Detailed Attraction View: Users should have the option to click on any attraction to discover more about it in detail, including the description of the attraction, historical importance, high-quality images, approximate time to reach the destination from the user's current location, fees to pay if any, and the hours of operation.

Bilingual Content Delivery: The system should have the capability to deliver all textual information in both Sinhala and English languages to the user in order to switch between these two languages smoothly.

Seamless Navigation: There should be a "Navigate" button on the detailed page of each attraction that should open Google Maps with a predefined route from the user's current location to the GPS coordinates of the chosen attraction.

Curated Day Trip Planning: The system should have "One Day Trip" options that are logically grouped together to help users plan their day more effectively.

2.3. User Classes and Characteristics

The following user classes have been identified for the "Village Traveler" system:

Independent Tourists: These are budget-conscious tourists who travel alone and rely on smartphone apps for information. These tourists require a user-friendly system that is fast and efficient with accurate information to make impromptu decisions.

Cultural and Eco Tourists: These tourists are individuals or couples or families who have a special interest in history, art, etc. These tourists require detailed and accurate descriptions of the tourist locations, their importance, and their surroundings in terms of culture and nature.

Domestic Weekend Tourists: These tourists are Sri Lankan citizens from urban areas such as Colombo who look to explore new weekend destinations. These tourists require information about new places to visit and how to reach those places.

System Administrator: This user class is responsible for maintaining the system. These users require access to the system to update the database containing information about tourist locations.

2.4. Operating Environment

The software will be a web application. As such, it will have to be compatible with the following environments:

Client Side (User): The software will have to be fully functional with the latest versions of major browsers like Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge. The software will have to be fully responsive with a focus on smartphones and tablets used by tourists during their travels.

Server Side (Hosting): The software will be hosted on a cloud platform with services like Firebase used for backend operations.

2.5. Design and Implementation Constraints

The constraints that will guide the design and implementation of this system are as follows:

Geographical Scope: The scope of this application is explicitly limited to a 25km radius around the center of Wathugedara.

Technology Stack: This application is expected to be built using the technologies discussed in Section 1.9 of the project proposal, which are as follows:

Frontend: React or Angular

Backend: Spring Boot

Database: Firebase (as stipulated in the objectives of this project)

Version Control: Git

Third-Party Services: The application is expected to use Google Maps API, which is subject to the terms and conditions and usage limits thereof.

Content Scope: The scope of this application is limited to 10 major tourist attractions, as stipulated in the project proposal.

2.6. User Documentation

The user documentation that will be provided with the software includes the following:

Integrated On-Screen Guidance: There will be a basic introduction for first-time users of the application on the basic features of the application.

About Section: There will be a special page or section in the application that will provide a brief overview of the project, its purpose, and the team involved.

2.7. Assumptions and Dependencies

Assumptions:

User Device Capabilities: It is assumed that users have a modern smartphone, tablet, or computer with GPS enabled on the browser and an active internet connection (4G or WiFi) to take full advantage of the system.

Google Account: It is assumed users have access to Google Maps, and consequently, a Google account on their device to enable the navigation feature.

Data Accuracy: It is assumed the data provided on the initial 10 attractions is accurate at the time of system launch.

Dependencies:

Google Maps API: The project depends on the availability, stability, and cost of the Google Maps Platform, including the JavaScript API, for the location display and potential integration of turn-by-turn navigation.

Firebase Services: The project depends on the stability and uptime of Google's Firebase platform.

3. External Interface Requirements

3.1. User Interfaces

General: The user interface will be clean, intuitive, and image-centric with high-quality images of the tourist attractions. The user interface will be designed in accordance with responsive web design principles to provide a consistent user experience across desktops, tablets, and mobile phones.

Language Toggle: A prominent feature for toggling between Sinhala and English will be provided on all pages containing textual content. Users will be able to toggle between both languages instantly without reloading the core page content.

Attraction Listing: The attractions will be provided in a clear listing or card view. The listing will be done in order of proximity to the user. For each attraction, the name of the attraction, a thumbnail image, category, and distance from the user will be provided.

Detail View: On clicking an attraction, a detailed view will be provided with images, text content for the attraction, and clear headings for the content (e.g., "Opening Hours," "Entrance Fee").

Error Handling: Friendly error messages will be provided for cases like GPS not working (e.g., "Unable to get your location. Please ensure location services are enabled.") or for cases like data loading failures.

3.2. Hardware Interfaces

- **GPS Receiver:** The system will interface with the device's built-in GPS receiver via the browser's Geolocation API to obtain the user's current latitude and longitude coordinates.

3.3. Software Interfaces

Google Maps JavaScript API: This API will be used to achieve the following:

Displaying an interactive map showing all the locations of the attractions.

Passing the current location of the user and the selected attraction's coordinates to the Google Maps app to achieve turn-by-turn directions.

Firebase (Backend as a Service): This API will be used to achieve the following:

CRUD operations on the data of the tourist attractions.

Retrieving the list of places, details, images, and other information.

Web Browser Geolocation API: This API will be used to achieve the following:

Requesting and retrieving the geographical location of the user.

3.4. Communications Interfaces

Protocol: The protocol to be used between the client (web browser) and the backend services (Firebase, Google Maps API) will be HTTPS protocol, which ensures the encryption, integrity, and security of the data that is to be transmitted.

Data Format: The exchange of data with Firebase will be in JSON format.

4. System Features

This section details the major functional features of the "Village Traveler" system, in priority order.

4.1. Location-Based Attraction Discovery and Distance Display

Priority: High

Description: This is the main feature of the program. This program will automatically identify the user's location, obtain the list of tourist attractions from the database, compute the distance from the user to the tourist

attraction, and display the list in order from closest to farthest from the user. This is directly in line with the objective in Section 1.3.

Stimulus/Response Sequences:

User Action: User opens "Village Traveler" website.

System Response: Browser requests permission from user to access their location.

User Action: User grants permission.

System Response: The program computes the distance from the user to all the attractions in the database and displays them in a list from closest to farthest.

4.2. Bilingual Information Presentation

Priority: High

Description: To serve both local and international tourists, all textual data in the application will be presented in both Sinhala and English, and the application will have a simple mechanism to switch between these two languages. This meets the objective mentioned in Section 1.3.

Stimulus/Response Sequences:

User Action: The user clicks on a language toggle button, such as "සිංහල" and "English."

System Response: The application interface and all textual data pertaining to the currently displayed attraction(s) will be dynamically changed to the chosen language without reloading the page.

4.3. Turn by Turn Navigation Integration

Priority: High

Description: This feature will serve as a seamless gateway between the "Village Traveler" application and Google Maps, enabling users to access turn-by-turn directions to their selected destination. This will be equivalent to meeting the project requirements as mentioned in Section 1.3.

Stimulus/Response Sequences:

User Action: The user selects a button labeled "Navigate" or "Get Directions" on a detailed view of an attraction.

System Response: The application will send a request to open the user's default Google Maps application or will open Google Maps in a new browser window with a request for directions from the user's current location to the attraction's coordinates.

4.4. One Day Trip Planning

Priority: Medium

Description: This feature is designed to help the user plan their trip by offering them a list of day trips, where groups of nearby attractions are organized in a logical and time-efficient way. This is described in the project overview (Section 1.7).

Stimulus/Response Sequences:

User Action: The user visits a "Day Trips" or "Itineraries" section of the website.

System Response: The system shows a list of predefined day trip plans (e.g., "Madu River Explorer," "Heritage and Gems Tour"), and clicking on any plan shows a route and a list of included attractions.

5. Other Nonfunctional Requirements

5.1. Performance Requirements

- The main application window and list of tourist attractions shall take less than 3 seconds to load over a normal 4G network connectivity, as defined under the project objectives (Section 1.3).
- Calculating distances and reordering the list shall take less than 2 seconds after obtaining user location information.
- Switching between language modes (Sinhala/English) shall happen in real-time with minimal delay (less than 1 second for text updates).

5.2. Security Requirements

The client and backend service (Firebase) communications shall be encrypted using HTTPS.

The access to administrative features such as adding, editing, or deleting attraction information shall be restricted using a secure form of authentication such as Firebase Authentication and shall be accessible only by System Administrators.

The application shall not store information regarding tourists such as location history and search history, which are considered PII.

5.3. Software Quality Attributes

Usability: The system shall be user friendly for first-time users. The basic operations of the system, like finding nearby places and viewing details, shall be possible in three taps or clicks.

Reliability: The system shall have an availability of 99.5% during the daytime (6:00 AM to 8:00 PM), as tourists will be using the system during this time. The system shall be able to handle errors from external sources (like Google Maps) without crashing.

Maintainability: The code shall be written in a manner that makes it easier for another developer to modify it in the future. Git shall be used for version control.

Availability: The system shall be available 24/7 except during scheduled maintenance.

5.4. Business Rules

The information system is purely informational. No commercial transactions, such as ticket bookings, hotel bookings, etc., will be carried out through this system.

Information related to tourist spots should always be accurate and factual. No promotional information should be displayed.

6. Other Requirements

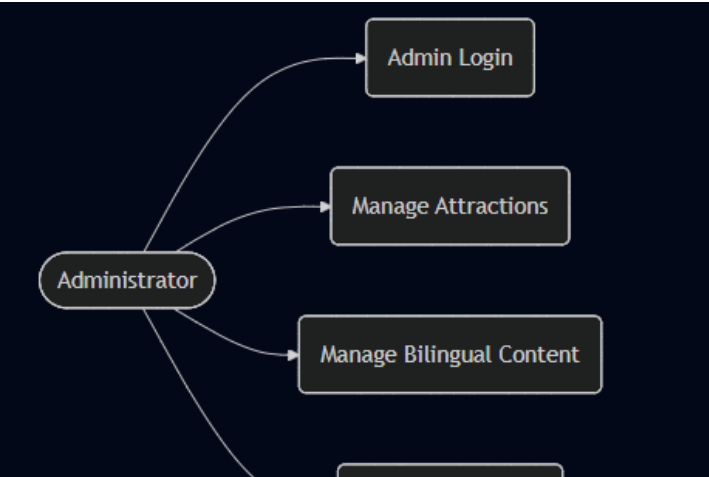
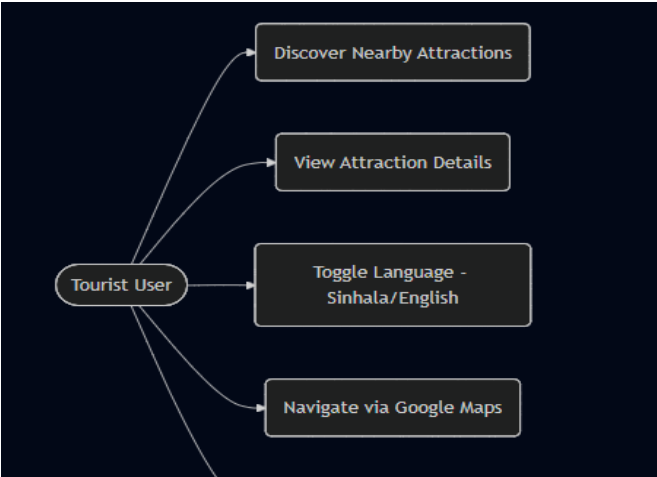
Internationalization: The initial product will be designed to accommodate two languages. The design of the system will be such that it will be easy to add more languages in the future & Develop a mobile app.

Appendix A: Glossary

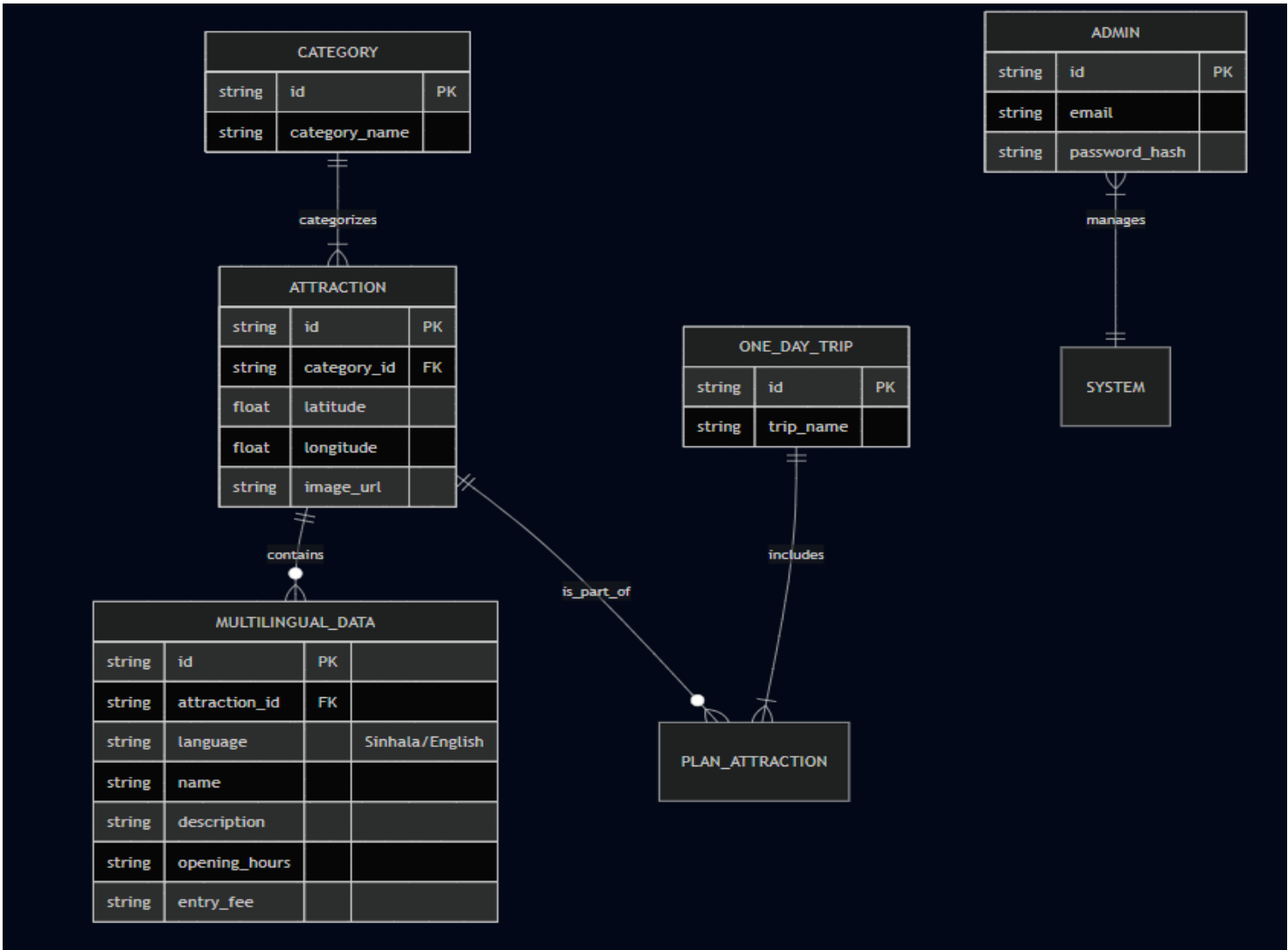
Term	Definition
API (Application Programming Interface)	A set of definitions and protocols for building and integrating application software, allowing two applications to talk to each other.
Bilingual	The use of two languages; in this project, English and Sinhala.
Firebase	A platform developed by Google for creating mobile and web applications, used here as the database and backend service.
GPS (Global Positioning System)	A satellite-based radio navigation system that provides geolocation and time information to a GPS receiver anywhere on Earth.
Geolocation API	A browser API that allows web applications to access the device's location.
IEEE	Institute of Electrical and Electronics Engineers.
SRS	Software Requirements Specification.
Turn by Turn Navigation	A feature of a GPS navigation device that gives spoken or visual directions to a driver as they approach each maneuver.

Appendix B: Analysis Models

Use Case Diagram



ER Diagram



High-level Picture Diagram

